

Ethics

PUBH 526

Design and Analysis of Randomized Trials in Public Health

Nilupa S. Gunaratna, MS, PhD

ASA Ethical Guidelines

- Intended for “statistical practitioners”, regardless of job title
- Intended to be actionable, to help make decisions
- Also informs those who rely on statistics and statistical practitioners of the standards they can expect from us and our profession
- Good statistical practice:
 - Transparent assumptions
 - Reproducible results
 - Valid interpretations

2022 Guidelines



Principle A: Professional Integrity and Accountability

4. Opposes efforts to predetermine or influence the results of statistical practices and resists pressure to selectively interpret data.

Principle B: Integrity of Data and Methods

1. Communicates data sources and fitness for use, including data generation and collection processes and known biases. Discloses and manages any conflicts of interest relating to the data sources. Communicates data processing and transformation procedures, including missing data handling.

Principle D: Responsibilities to Research Subjects, Data Subjects, or Those Directly Affected by Statistical Practices

5. Uses data only as permitted by data subjects' consent, when applicable, or considers their interests and welfare when consent is not required. This includes primary and secondary uses, use of repurposed data, sharing data, and linking data with additional data sets.
6. Considers the impact of statistical practice on society, groups, and individuals. Recognizes that statistical practice could adversely affect groups or the public perception of groups, including marginalized groups. Considers approaches to minimize negative impacts in applications or in framing results in reporting.

ASA Ethical Guidelines – Principles

- One principle may raise an ethical issue, and other principle may provide a solution
- Sometimes principles may conflict – prioritization of principles depends on context
- Goal: advance knowledge while avoiding harm?
- You will encounter ethical issues, especially if you work with disadvantaged populations or under limited resources
- You may often be the only statistician on the team (whether or not that is your job title)

Ethical Reasoning

- Identify and assess your prerequisite knowledge
 - When in doubt, seek help
- Recognize an ethical issue (a decision to be made)
- Identify relevant decision-making frameworks
 - Prioritize minimizing harm, prioritize obligations to profession, ...
- Identify and evaluate alternative actions
 - “No action” is an action!
- Make and justify a decision
- Reflect on the decision

ASA Statement on Ethical AI

Principles for Statistical Practitioners

- **Accountability Principle:** Statistical practitioners and their organizations should be capable of auditing and justifying the AI systems they use or help develop.
- **Transparency Principle:** Statistical practitioners and their organizations should promote transparency to build trust in the AI systems they use or help develop.
- **Fairness Principle:** Methods and technologies should be developed in ways that promote fairness and equity among those affected by the use of AI technologies.

Principles and Practices for a Federal Statistical Agency

“The Purple Book”

Through the director of the Office of Management and Budget (OMB), the Chief Statistician of the United States has the responsibility of coordinating domestic and international federal statistical policy (Foundations for Evidence-Based Policymaking Act of 2018, 2019; Paperwork Reduction Act, 1995).

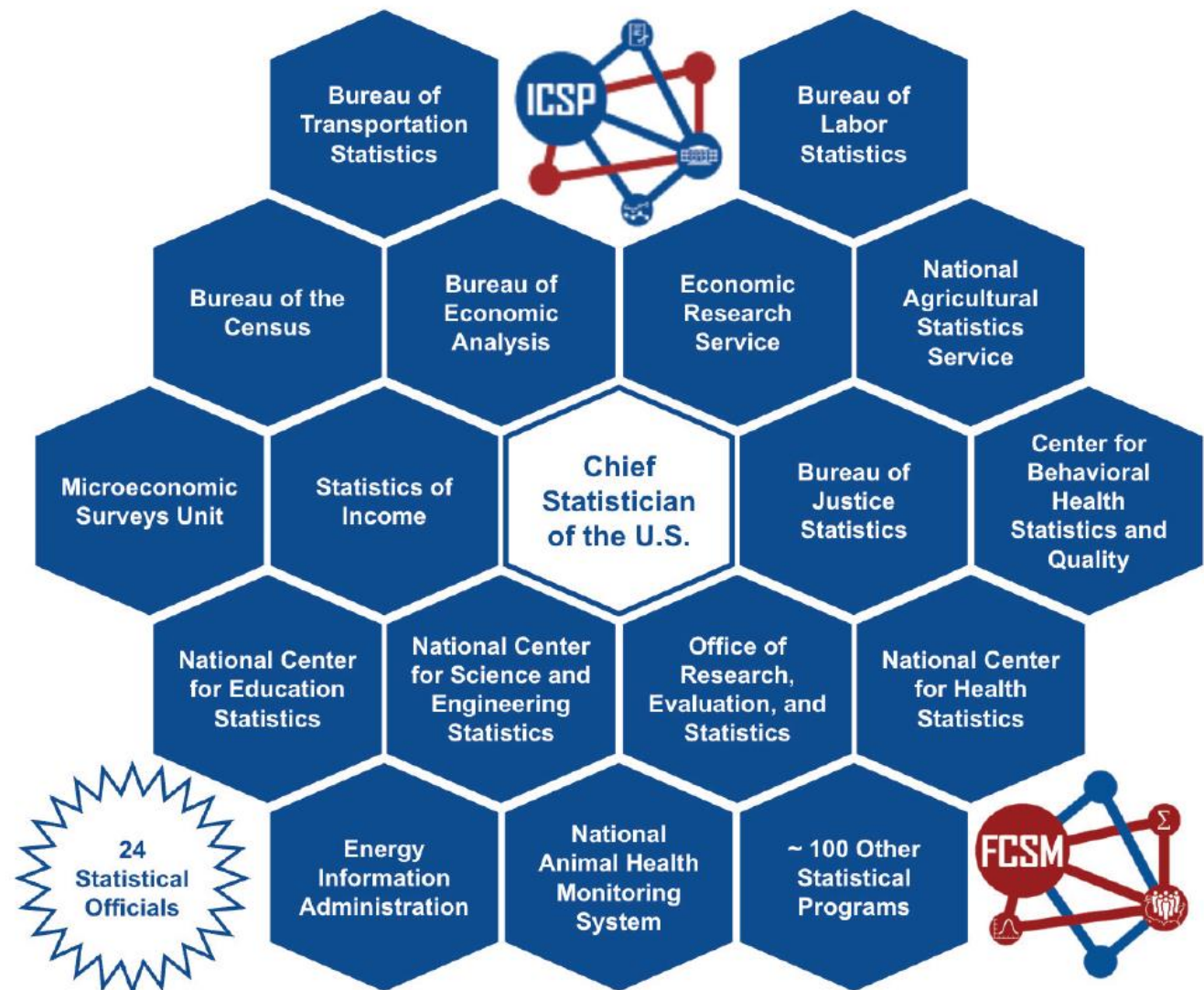


FIGURE 1-1 The U.S. federal statistical system as depicted on StatsPolicy.gov.
SOURCE: <https://www.statspolicy.gov/about/#office-chief>

BOX S-1

Principles and Practices

Five principles guide federal statistical agency programs:

1. Relevance to Policy Issues and Society
2. Credibility Among Data Users and Stakeholders
3. Trust Among the Public and Data Subjects
4. Independence from Political and Other Undue External Influence
5. Continual Improvement and Innovation

Ten practices support achievement of these principles:

1. A Clearly Defined and Well-Accepted Mission
2. Necessary Authority and Procedures to Protect Independence
3. Commitment to Quality and Professional Standards of Practice
4. Professional Advancement of Staff
5. An Active Research Program
6. Strong Internal and External Evaluation Processes for an Agency's Statistical Programs
7. Coordination and Collaboration with Other Agencies
8. Respect for Data Subjects and Data Holders and Protection of Their Data
9. Dissemination of Statistical Products That Meet Users' Needs
10. Openness About Sources and Limitations of the Data Provided

Federal Register :: Statistical Policy

federalregister.gov/documents/2014/12/02/2014-28326/statistical-policy-directive-no-1-fundamental-responsibilities-of-federal-statistical-ag...

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 **FEDERAL REGISTER**
The Daily Journal of the United States Government 

Notice

Statistical Policy Directive No. 1: Fundamental Responsibilities of Federal Statistical Agencies and Recognized Statistical Units

A Notice by the Management and Budget Office on 12/02/2014

PUBLISHED DOCUMENT: 2014-28326 (79 FR 71610)

PDF
Document Details
Document Dates
Table of Contents
Public Comments
Regulations.gov Data
Sharing
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Document Statistics
Other Formats
Public Inspection

DOCUMENT HEADINGS
Office of Management and Budget

AGENCY:
Executive Office of the President, Office of Management and Budget.

ACTION:
Notice of final decision.

SUMMARY:
Under the *Budget and Accounting Procedures Act* of 1950 (31 U.S.C. 1104 (d)) and the *Paperwork Reduction Act* of 1995 (44 U.S.C. 3504 (e)), the Office of Management and Budget (OMB) is issuing Statistical Policy Directive No. 1, *Fundamental Responsibilities of Federal Statistical Agencies and Recognized Statistical Units*. This Directive affirms the fundamental responsibilities of Federal statistical agencies and recognized statistical units in the design, collection, processing, editing, compilation, storage, analysis, release, and dissemination of statistical information. On May 21, 2014, OMB published a Notice of solicitation of comments on a draft of this Directive in the **Federal Register** (79 FR 29308, May 21, 2014). Eight respondents sent comments in regard to the notice. Careful consideration was given to all comments. The disposition of the comments as well as the final Directive are presented in the **SUPPLEMENTARY INFORMATION** section below.

In its role as coordinator of the Federal statistical system under the *Paperwork Reduction Act*, OMB, among other responsibilities, is required to ensure the efficiency and effectiveness of the system as well as the integrity, objectivity, impartiality, utility, and confidentiality of information collected for statistical purposes. OMB is also charged with developing and overseeing the implementation of Government-wide principles, policies, standards, and guidelines concerning the development, presentation, and

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“It is the responsibility of Federal statistical agencies and recognized statistical units to produce and disseminate relevant and timely information; conduct credible, accurate, and objective statistical activities; and protect the trust of information providers by ensuring confidentiality and exclusive statistical use of their responses”

How Is This Achieved?

- Other Statistical Policy Directives
 - For example, SPD 2 governs the release of principal federal economic indicators (e.g., unemployment rate, CPI, GDP)
 - Public release schedules announced in advance
 - Simultaneous public access (no early access by policymakers or markets)
 - Strict embargo and security procedures to prevent leaks

How Is This Achieved?

- Separation of roles: Political appointees often oversee budgets, policy alignment, and interagency coordination, but career staff retain authority over:
 - Sampling methods
 - Questionnaire design
 - Data analysis and estimation
 - Statistical disclosure decisions
- External Oversight: Congressional committees, the Government Accountability Office (GAO), and others

How Is This Achieved?

- Evidence Act of 2018 (Foundations for Evidence-Based Policymaking Act of 2018)
 - Promotes evidence-based policymaking
 - Improves transparency and accountability
 - Strengthens data governance and privacy/confidentiality protections
- Confidential Information Protection and Statistical Efficiency Act (CIPSEA)
 - Balances data sharing with privacy and confidentiality
 - Establishes mechanisms for secure data linkage
 - Restricts the use of data for non-statistical purposes (enforcement, regulation, taxation, immigration, policy targeting, etc.)

Assessing the Health of the
Federal Statistical Agencies

The Nation's Data at a Crossroads

Year Two Status Report

July 31, 2025; amended September 3, 2025*